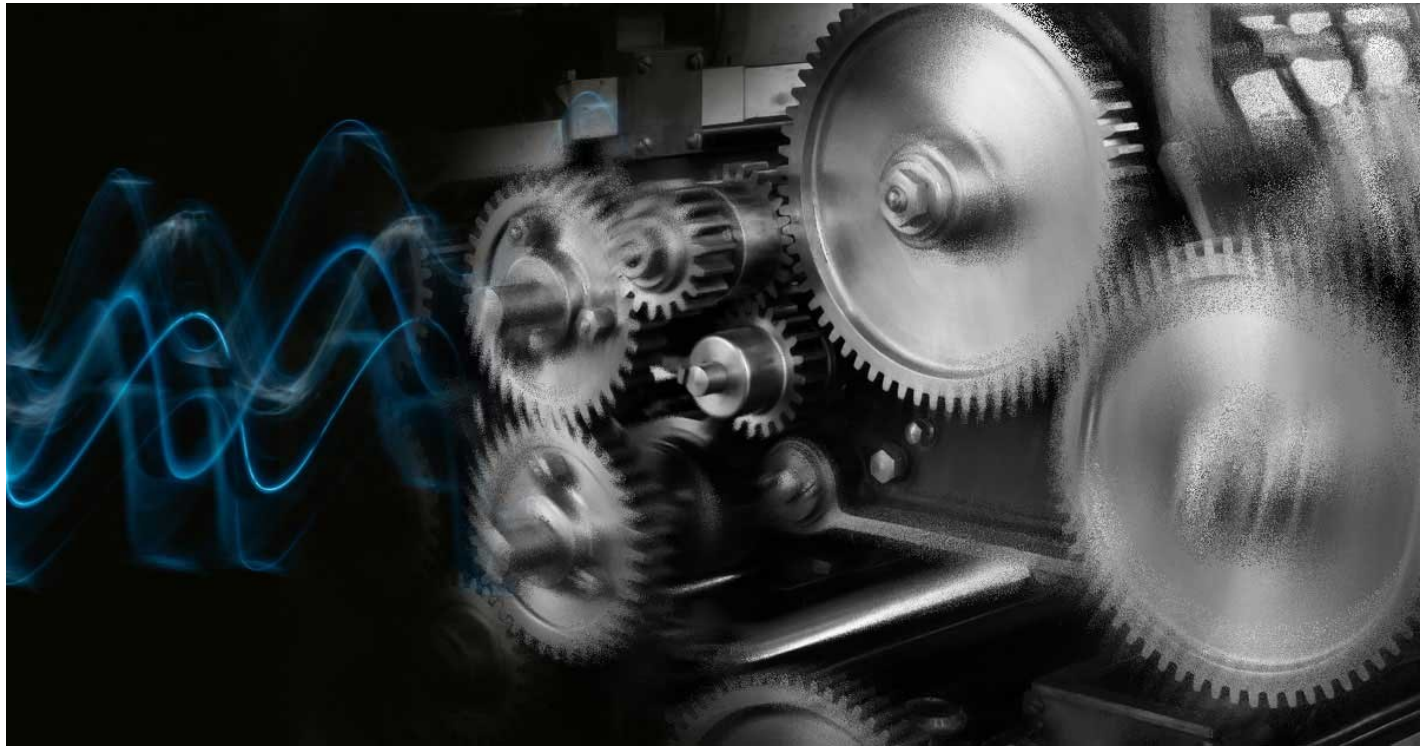
Challenge 1

MechPulse – Smart Vibration Monitoring for Industrial Reliability

Challenge 1: MechPulse

Challenge Overview

Vibration is a key indicator of mechanical health in industrial equipment. It reflects the dynamic behaviour of rotating and moving parts, and deviations from normal vibration patterns often indicate wear, imbalance, misalignment, or impending mechanical failure. Monitoring vibration is essential for maintaining operational safety, reducing downtime, and extending equipment lifespan. Despite its importance, many systems still rely on manual inspections or periodic measurements, which may miss early warning signs. Vibration analysis is widely used in predictive maintenance but implementing it effectively across diverse machinery and environments remains a challenge.



Challenge 1: MechPulse

Scope and Requirements

Industrial machinery operates under varying loads, speeds, and environmental conditions, making early fault detection a complex but critical task. Traditional vibration monitoring systems often rely on fixed sensor configurations and centralized data analysis, which may not be suitable for dynamic or resource-constrained settings. Moreover, many existing solutions are expensive and lack flexibility, limiting their accessibility and scalability.

You are invited to design a smart vibration monitoring system that continuously collects and analyses vibration data to detect anomalies early and support predictive maintenance. The system should be adaptable to diverse industrial environments. The solution should aim to improve reliability, safety, and efficiency, while being robust, user-friendly, and capable of offering real-time insights to maintenance teams.



Challenge 2

CorroSense – Non-Invasive Corrosion Monitoring in Sealed Metallic Enclosures

Challenge 2: CorroSense

Challenge Overview

Corrosion is a slow but destructive process that affects the structural integrity and performance of metallic systems. In sealed metallic enclosures—used in aerospace, marine, and industrial applications—corrosion can develop due to trapped moisture, temperature fluctuations, or chemical exposure. These environments are often inaccessible, making it difficult to monitor internal conditions without compromising the enclosure's integrity. Such enclosures are designed to protect sensitive electronics or mechanical components from harsh external conditions, and any breach can introduce vulnerabilities. Over time, undetected corrosion can lead to system failures, safety hazards, and costly maintenance. Monitoring corrosion within sealed metallic enclosures presents unique challenges due to the need for environmental isolation and non-invasive data transmission.

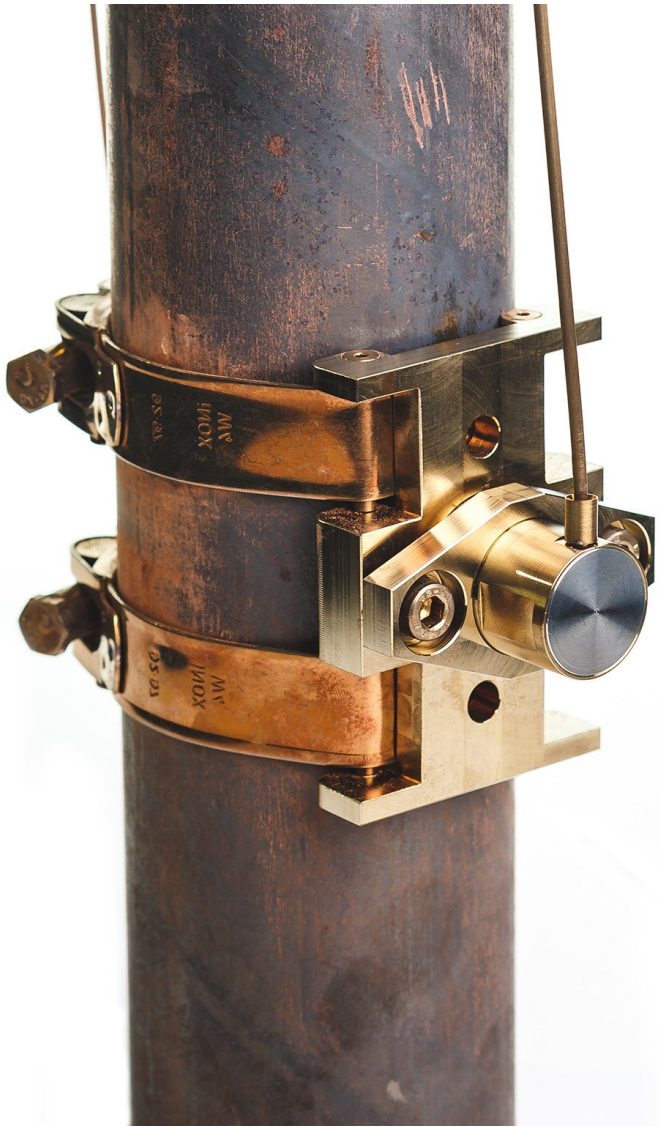


Challenge 2: CorroSense

Scope and Requirements

Corrosion is influenced by multiple factors, including humidity, temperature, and the presence of corrosive agents, all of which must be monitored to assess risk accurately. Conventional approaches using wired sensors and feedthroughs compromise the enclosure's integrity and are not suitable for long-term deployment. To overcome these limitations, wireless sensing technologies offer a promising alternative by enabling continuous monitoring without breaching the protective barrier of the enclosure.

You are challenged to design a system that monitors corrosion-related conditions—such as humidity, temperature, and chemical markers— inside a closed metallic enclosure and communicates data externally without using electrical feedthroughs. The goal is to enable early detection and intervention while preserving the enclosure's protective function and long-term reliability. This system must operate autonomously over extended periods, withstand harsh environmental conditions, and maintain minimal power consumption to ensure feasibility for remote or inaccessible installations.



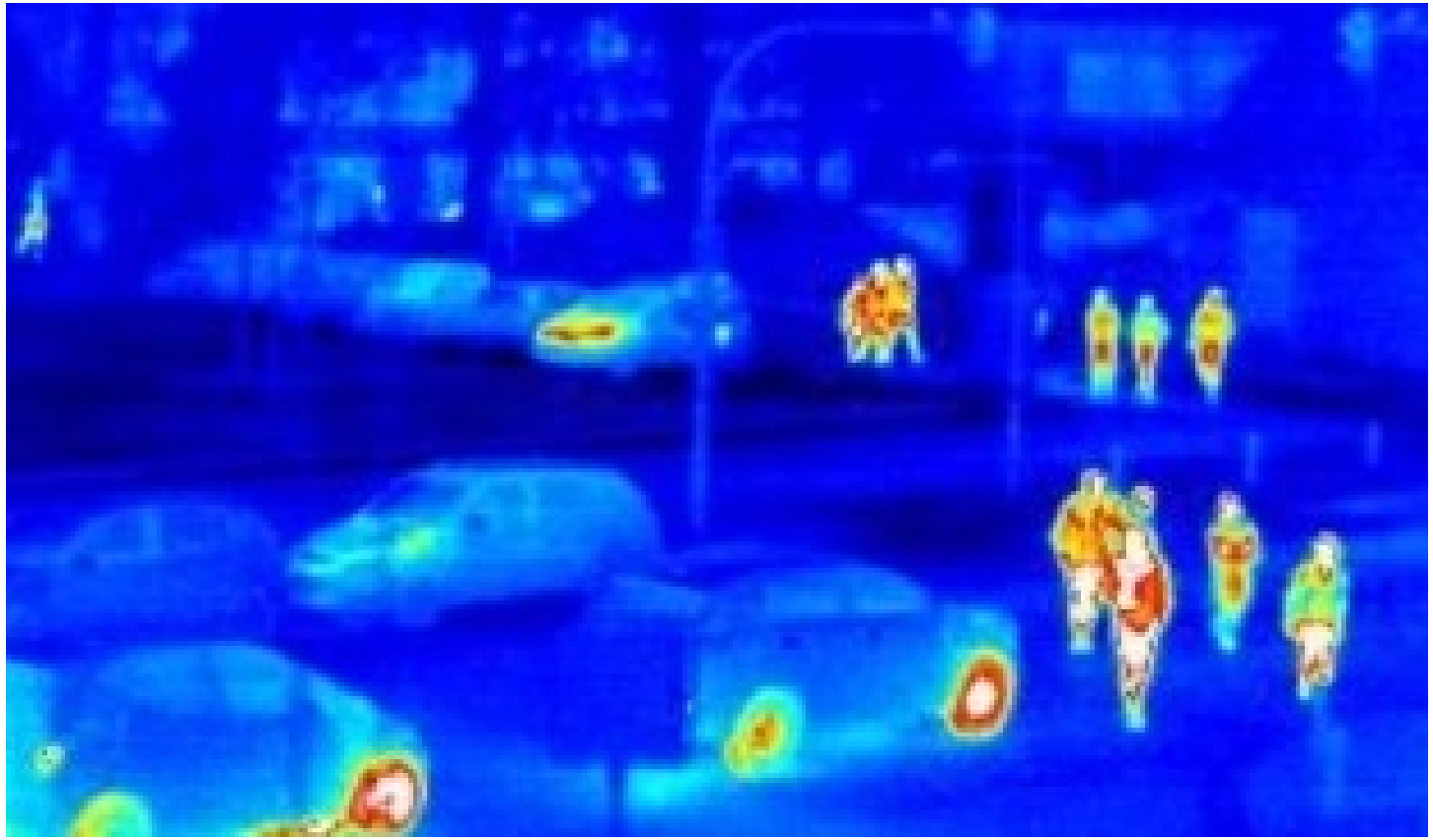
Challenge 3

Night-Drive – Enhanced Detection of Road Hazards Beyond Visible Light

Challenge 3: Night-Drive

Challenge Overview

Night driving poses significant safety challenges due to the inherent limitations of standard vehicle headlights. Conventional headlights provide a restricted field of view and limited illumination distance, leaving drivers vulnerable to hazards beyond the beam's reach. Additionally, glare from oncoming traffic can further reduce visibility, increasing the risk of accidents. These limitations contribute to a disproportionate number of nighttime collisions involving pedestrians, animals, and other vehicles. Despite advancements in automotive lighting and driver-assistance systems, achieving reliable detection in low-light conditions remains a critical challenge for road safety.

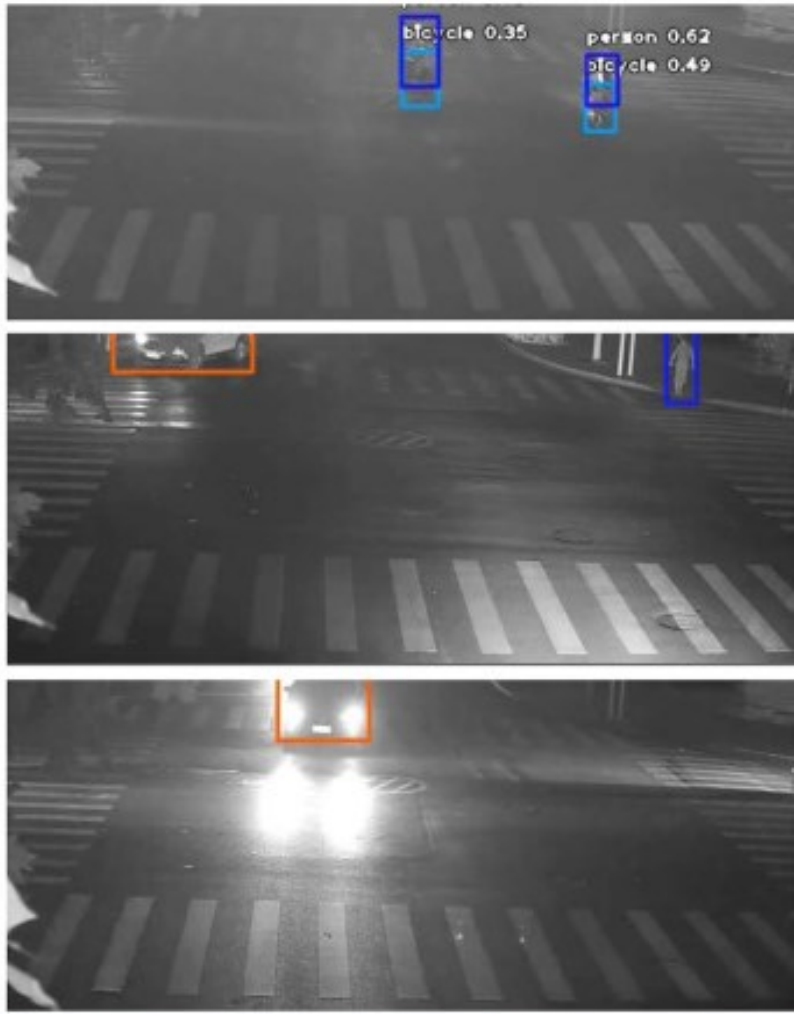


Challenge 3: Night-Drive

Scope and Requirements

Using normal car headlights for night driving has its limitations, such as limited field of view, limited illuminated distance, and dazzling from oncoming traffic, to name a few. Thus, a lot of accidents and collisions happen during night driving—with pedestrians, with other cars, and with animals on the road. Traditional vision-based systems that rely on visible light struggle in these conditions, especially in rural or poorly lit environments.

You are invited to propose a solution that does not depend on normal visible lights and achieves pedestrian detection using passive or active imaging technologies operating in wavebands or modalities beyond the visible spectrum (e.g., infrared, thermal, radar, lidar, or ultrasound). The system should detect the presence of pedestrians, animals, and other vehicles in real time and react to avoid collisions. The solution must be robust under varying weather conditions, energy-efficient for automotive integration, and capable of providing actionable alerts or automated responses to enhance driver safety.



Challenge 4

CyanoDetect – On-Site Real-Time Detection of Cyanotoxin Microcystin on Vegetables

Challenge 4: CyanoDetect

Challenge Overview

Microcystin is a highly toxic compound produced by blue-green algae (cyanobacteria) during algal blooms in nutrient-rich waters. It poses serious health risks to humans and animals, including liver damage when exposed to contaminated water. In Tønsberg municipality, Microcystin is detected annually in Lake Akersvannet, and several incidents of animal poisoning have been reported including death of several dogs after drinking water from the lake. The concern extends to agriculture, where irrigation water contaminated with this cyanobacterial toxin may lead to microcystin contamination of vegetables. This creates a potential food safety hazard and threatens consumer health. Current laboratory-based detection methods are accurate but slow, costly, and unsuitable for on-site monitoring. There is an urgent need for a rapid, reliable, and field-deployable solution to ensure food safety.



Challenge 4: CyanoDetect

Scope and Requirements

Tønsberg municipality needs equipment to detect the amount of the cyanotoxin Microcystin on the surface of vegetables after irrigation with contaminated water. The municipality wants to monitor Microcystin levels in vegetables on-site in real time without relying on centralized laboratory testing. The innovative solution should integrate a biosensor capable of detecting Microcystin and provide immediate, actionable results to farmers and authorities.

You are invited to design a portable, user-friendly system for on-site detection of Microcystin on vegetables. The system should be accurate, robust, and capable of operating under field conditions. It must deliver real-time results, minimize sample preparation, and ensure low power consumption for practical deployment. The solution should also consider scalability and cost-effectiveness to enable widespread adoption in agricultural monitoring.



Challenge 5

AutoSwab – Automated Sampling for ENT Cancer and Respiratory Pathogen Detection

Challenge 5: AutoSwab

Challenge Overview

In relation to the prevention and control of pandemics, epidemics, and cancer related to respiratory diseases—including 23 different viruses and bacteria—and associated with several cancer types in the area from the laryngopharynx to the upper tonsil, early detection is critical. Globally, head and neck cancers are estimated to cause around 487,993 deaths annually according to the 2020 Global Cancer Statistics, though this figure can vary by source and year.

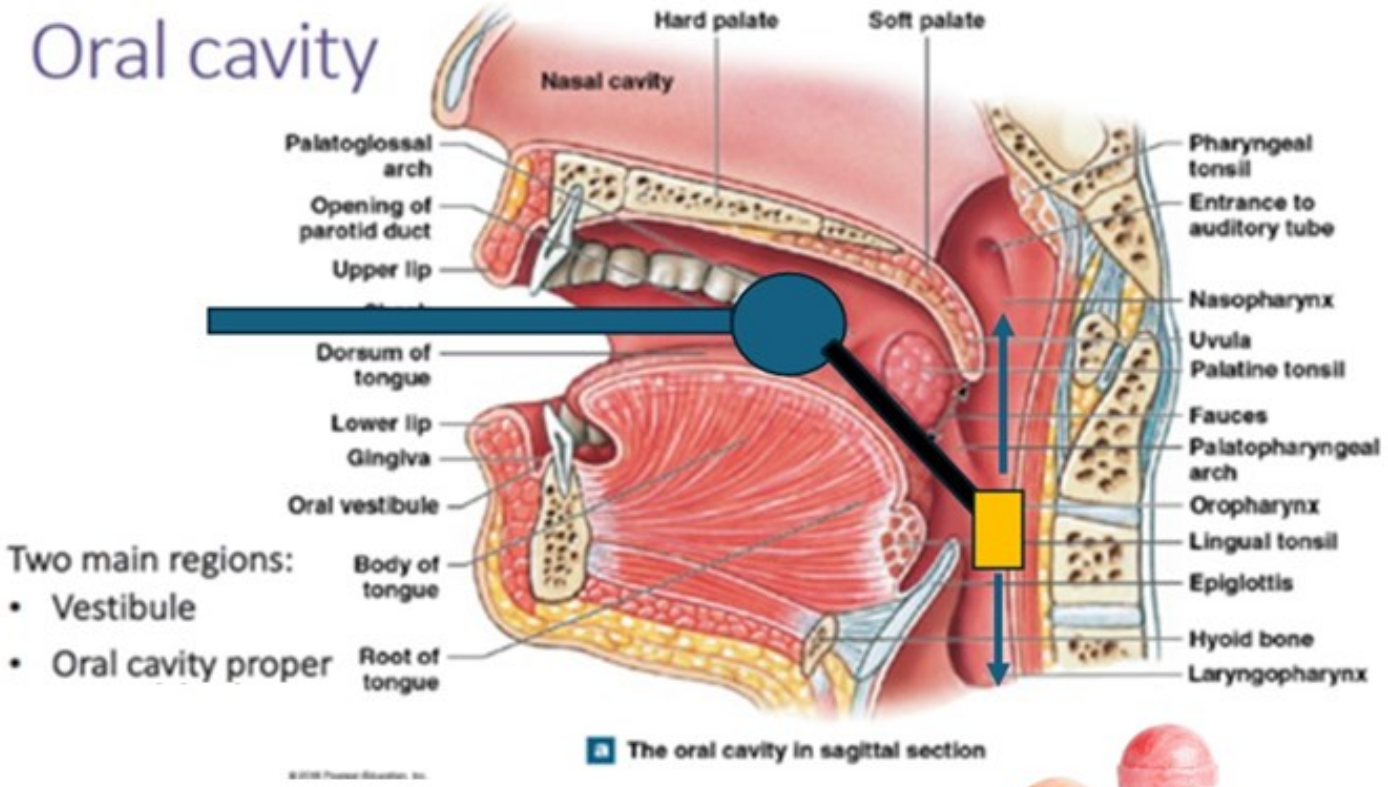
Today, we have a number of biosensor technologies, including in-house solutions, that can directly identify active viruses, bacteria, and pre-cancer transcript profiles. These technologies are ideal for use within national screening programs and can cover any high-risk population. The challenge is to develop a new way of taking representative samples from the ENT (Ear, Neck, and Throat) area in the most automatic and user-friendly way, suitable for all ages.



Challenge 5: AutoSwab

Scope and Requirements

The suggested Lollipop method may be the best way to collect samples from the whole representative area within seconds and move the sample into the head of the Lollipop. The surface of the brush used must be ideal for collecting all kinds of biological particles or cells inside saliva/sputum samples. The mechanical unit and devices must be stored inside the Lollipop head, and innovations have to be made to make it quickly move automatically out of the head and in again. The user may control when the collection process starts and when it is finished.



The brush must consist of a rolling unit that can very quickly move up and down the throat area. After the Lollipop has been used, it must be transferred to a transport container with RNA- and DNA-friendly media. In case it is used by children, it is wise to include a normal Lollipop as a replacement possibility. This kind of Lollipop technology should be able to be used at home and may become one of the best technologies for the control of ENT cancer and all kinds of respiratory-related pandemics or epidemics.



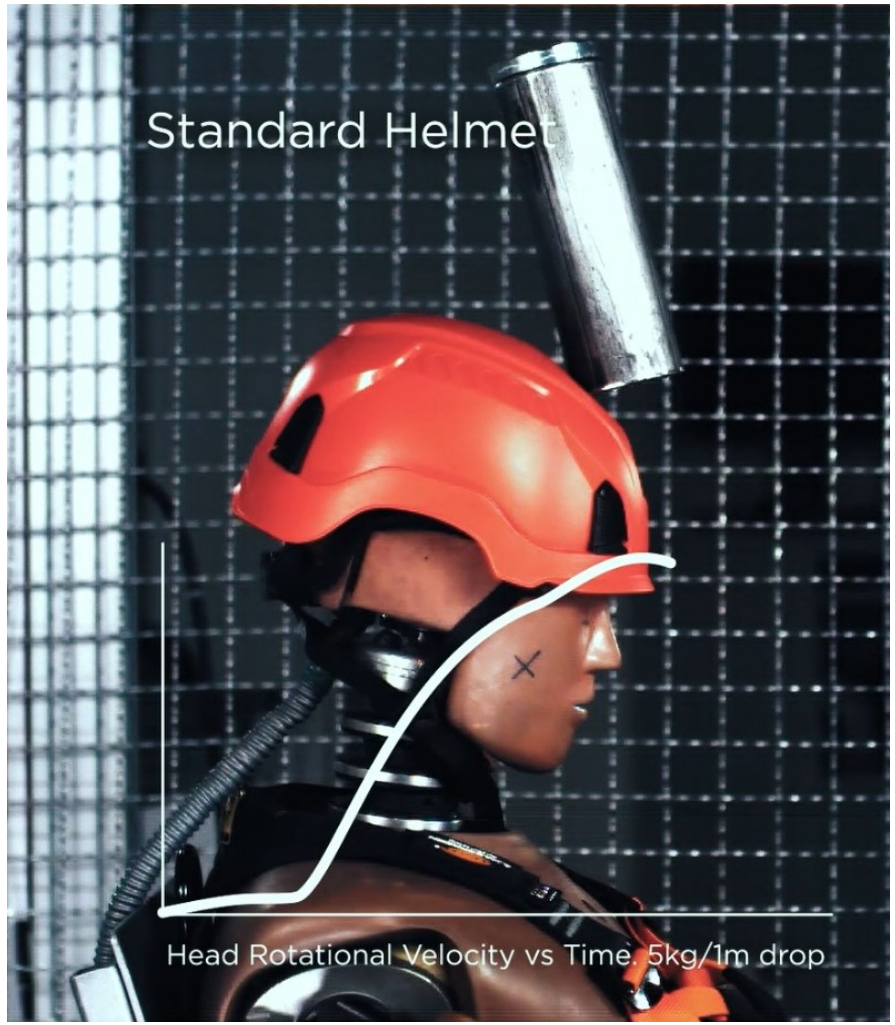
Challenge 6

SmartGuard – Smart Wearables for Impact and Fall Detection

Challenge 6: SmartGuard

Challenge Overview

Accidents in construction, cycling, industrial work, and emergency response environments pose significant risks to human safety. Head injuries, falls, and impact-related accidents are among the leading causes of severe injuries and fatalities in these sectors. In many cases, the lack of immediate medical attention following an accident exacerbates the severity of injuries. Moreover, workers often operate in isolated or hazardous conditions where timely communication is difficult, increasing the risk of delayed response. Similarly, elderly individuals are highly vulnerable to falls, which can lead to long-term health complications or death. Despite advancements in personal protective equipment, most helmets and safety gear lack integrated systems for real-time accident detection and alerting. This gap in functionality limits the effectiveness of safety measures, especially in dynamic or high-risk environments. There is a growing need for smart, sensor-enabled solutions that can monitor impacts, detect falls, and provide timely alerts to prevent severe outcomes.



Challenge 6: SmartGuard

Scope and Requirements

Current protective gears provide physical safety but lacks intelligent monitoring and real-time alerting capabilities. Traditional solutions do not offer automated detection or communication features, which are critical for rapid emergency response. Integrating MEMS-based sensors such as accelerometers, gyroscopes, and pressure sensors into wearable devices can enable accurate detection of impacts and sudden falls across multiple scenarios.

You are invited to design a smart wearable system—such as a helmet or accessory—that can detect accidents in real time and communicate alerts effectively. The solution should be lightweight, robust, and capable of operating in harsh environments. It must ensure low power consumption, wireless connectivity, and user comfort, while being adaptable for diverse applications such as construction helmets, firefighter safety trackers, and fall detection for elderly patients. The goal is to enhance personal safety through intelligent, automated monitoring that reduces response time and saves lives.

